

Swati Sutcliffe

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PROFESSIONAL SUMMARY

Senior Data Engineer with 6+ years of experience designing and implementing enterprise-scale data solutions across healthcare, financial services, and insurance. Expertise in architecting metadata-driven ETL/ELT pipelines and lakehouse architectures using Azure Data Factory, Databricks, Spark, SQL, and Python. Proven ability to integrate structured, semi-structured, and streaming sources into secure, governed platforms in Microsoft and AWS environments.

TECHNICAL SKILLS

- **Cloud Platforms:** AWS (S3, Redshift, Glue, Lambda, Kinesis Data Streams, Firehose, EMR, Athena, RDS, DynamoDB, IAM, Lake Formation, CloudFormation, Step Functions, API Gateway, Secrets Manager, SNS, SQS, CloudWatch), Azure (Data Factory, Synapse Analytics, Data Lake Storage Gen2, Databricks, Event Hubs, Azure Functions, Key Vault, Purview, Azure Monitor), Google Cloud Platform (GCS, BigQuery, Dataflow, Pub/Sub, Dataproc), ADLS
- **Infrastructure as Code & DevOps:** Terraform, AWS CDK, AWS CloudFormation, Docker, Kubernetes (EKS/AKS/GKE), Helm, Git, GitHub, Bitbucket, Jenkins, GitHub Actions, Azure DevOps, CI/CD Pipelines, Automated Deployments
- **Data Engineering & ETL:** Apache Spark, PySpark, AWS Glue (Jobs, Crawlers), AWS Step Functions, Lambda-based ETL, Azure Data Factory, Databricks, Apache Airflow, Prefect, dbt (Data Build Tool), Apache Kafka, Apache Flink, AWS Kinesis, Batch & Streaming Pipelines, Data Ingestion Frameworks, Data Management, Data Orchestration
- **Programming & Scripting:** Python, SQL, PySpark, Bash, Shell Scripting, UNIX/Linux, Java
- **Data Warehousing & Modeling:** Snowflake, Amazon Redshift, Azure Synapse Analytics, Google BigQuery, Star Schema, Snowflake Schema, Dimensional Modeling, Normalization, Partitioning Strategies, Clustering, Materialized Views, Lakehouse Architectures (Delta Lake, Redshift Spectrum), Azure SQL Database
- **Big Data Ecosystem:** Hadoop, Hive, HBase, Pig, Spark, Databricks, Distributed Data Processing
- **Monitoring & Observability:** Amazon CloudWatch, AWS X-Ray, Azure Monitor, Log Analytics, Datadog, Prometheus, Grafana, Custom Metrics, Alerts, Pipeline Health Monitoring, SLA Tracking
- **Security, Governance & Compliance:** AWS IAM, KMS, VPC Endpoints, S3 Bucket Policies, Lake Formation, Glue Data Catalog, RBAC, Azure Key Vault, CloudTrail, Data Lineage, Data Cataloging (Azure Purview), HIPAA, GDPR, Data Security Architecture
- **Streaming & Messaging:** Apache Kafka, AWS MSK, AWS Kinesis (Streams & Firehose), Azure Event Hubs, Google Pub/Sub
- **Data Formats & Serialization:** Parquet, JSON, CSV, Avro, ORC, XML, Delta Lake
- **Data Quality & Validation:** Great Expectations, AWS Glue DataBrew, Deequ, Schema Drift Detection, Data Validation Frameworks
- **Visualization & Analytics Integration:** Amazon QuickSight, Tableau, Power BI, Looker, Mode Analytics, Google Data Studio, Athena / Synapse / BigQuery as BI backends
- **Machine Learning & AI Data Support:** MLflow, SageMaker Integration, Feature Engineering Pipelines, TensorFlow & Scikit-learn (basic), Data Pipelines for ML Models, LangChain (RAG Pipelines)
- **APIs & Metadata Management:** RESTful APIs, API Gateway, Glue Crawlers, Glue Catalog, Lake Formation, Azure Purview, Metadata-driven ETL, AtScale
- **Project Management & Collaboration:** Agile (Scrum), Jira, Confluence, ServiceNow, Sprint Planning, Cross-functional Stakeholder Collaboration, Technical Documentation

PROFESSIONAL EXPERIENCE

LifePoint Health

Senior Data Engineer

Jun 2024 - Present

Brentwood, TN

- Led the architecture and implementation of an enterprise Lakehouse data platform on AWS S3 with Parquet and Delta-based storage, organizing healthcare datasets into raw, curated, and trusted zones to support analytics across LifePoint's clinical operations, finance, and population health teams, incorporating data management best practices.
- Designed scalable data ingestion frameworks using AWS Glue, Lambda, and API Gateway to capture healthcare datasets from EHR systems, patient surveys, and third-party providers in FHIR and HL7 formats, enabling ingestion of 15M+ records into the centralized data lakehouse.
- Engineered distributed data processing pipelines using PySpark on AWS Glue and Databricks, transforming patient encounters, billing transactions, and operational data into standardized datasets used by the Clinical Analytics and Hospital Operations teams.
- Built event-driven streaming pipelines using Amazon Kinesis, GCP Pub/Sub, and Lambda, enabling near real-time processing of patient telemetry and operational monitoring data used by LifePoint's Digital Health and Care Quality teams.
- Architected the enterprise data storage layer using AWS S3 Lakehouse architecture, integrating Redshift Spectrum and BigQuery federated queries to enable analytics teams to query petabyte-scale healthcare datasets without duplicating storage across environments.
- Designed orchestration frameworks using AWS Step Functions, Apache Airflow, and GCP Cloud Composer, coordinating multi-stage ETL pipelines across ingestion, transformation, validation, and analytics layers supporting hospital performance reporting and enabling robust data orchestration.

- Collaborated with Clinical Data Science, Revenue Cycle Analytics, and Population Health teams to build patient outcome dashboards and operational KPIs powered by Amazon Redshift, BigQuery, and Snowflake analytics models.
- Led the migration of legacy ETL pipelines from on-premise systems to cloud-native pipelines using AWS Glue, Snowflake, and Redshift, reducing ETL runtimes and improving operational stability for enterprise healthcare reporting systems.
- Implemented enterprise data governance frameworks using AWS Lake Formation, IAM, and GCP IAM, enabling secure cross-department data sharing while ensuring a robust data security architecture with role-based access control and encryption policies for patient data.
- Built scalable transformation layers using dbt, SQL, and PySpark, creating reusable dimensional models for patient encounters, hospital utilization metrics, and claims analytics used across finance and operational reporting teams.
- Optimized analytical workloads by redesigning query strategies in Amazon Redshift and GCP BigQuery, implementing partitioning, clustering, and columnar storage strategies that reduced cloud compute costs by over \$200K annually.
- Implemented automated data quality validation frameworks using Great Expectations and AWS Glue DataBrew, validating schema integrity and identifying data anomalies before they impacted downstream clinical reporting systems.
- Developed monitoring and observability frameworks using Amazon CloudWatch, Datadog, and GCP Cloud Monitoring, building operational dashboards that track pipeline health, ingestion latency, and data platform performance across environments.
- Worked within an Agile SCRUM data platform team, managing development tasks through Jira and Confluence, participating in sprint planning, architecture reviews, and cross-team coordination with data scientists, BI engineers, and DevOps teams.
- Mentored junior engineers and led internal knowledge sessions on cloud-native Lakehouse architecture, pipeline design patterns, and healthcare data compliance, contributing to the growth and technical maturity of the LifePoint data engineering team.

Western Union

Jan 2022 - May 2024

Data Engineer

Milwaukee, WI

- Helped design and implement data ingestion pipelines using Azure Data Factory (ADF) to capture global remittance transactions, customer activity logs, and partner settlement files into Azure Data Lake Storage Gen2 (ADLS), supporting analytics for the Payments Analytics and Compliance Reporting teams.
- Built distributed ETL pipelines using PySpark on Azure Databricks, transforming large-scale remittance transaction data into structured datasets used by fraud detection analysts and customer behavior analytics teams.
- Assisted in building a Lakehouse architecture using Delta Lake on Azure Databricks, enabling ACID-compliant datasets and versioned financial transaction data that supported downstream reporting and machine learning workflows.
- Developed orchestration workflows using Azure Data Factory pipelines, coordinating ingestion, transformation, and loading of financial datasets into Azure Synapse Analytics and Azure SQL Database for enterprise reporting and Power BI dashboards.
- Collaborated with the Fraud Risk, Compliance, and Business Intelligence teams to develop data models supporting global remittance monitoring dashboards and regulatory reporting pipelines.
- Built streaming ingestion pipelines using Azure Event Hubs and Stream Analytics, enabling near real-time processing of cross-border transaction events for fraud monitoring and operational analytics.
- Assisted with migration of legacy AWS-based ingestion pipelines using Amazon S3 and AWS Glue into Azure Data Factory and Azure Data Lake, helping the organization consolidate data engineering workloads onto the Azure analytics platform.
- Supported migration of on-prem SQL Server reporting workloads to Azure Synapse Analytics, improving scalability for financial reporting datasets used by enterprise analytics teams.
- Implemented data cataloging and governance processes using Azure Purview, enabling metadata lineage tracking and improved dataset discoverability across the organization's analytics platform.
- Developed CI/CD workflows using Azure DevOps and Git, enabling automated deployment of ADF pipelines and Databricks notebooks across development and production environments.
- Assisted with implementing data quality validation frameworks using Great Expectations and Deequ, ensuring transaction datasets met compliance and reporting accuracy standards.
- Built monitoring dashboards using Azure Monitor and Log Analytics to track pipeline failures, ingestion delays, and data platform health across the Azure environment.
- Worked with the Data Science team using MLflow on Azure Databricks, helping operationalize customer analytics and fraud detection models into production pipelines.
- Integrated partner datasets stored in AWS S3 using Azure Data Factory connectors, enabling cross-cloud ingestion of partner transaction files into the Azure lakehouse environment.
- Worked within an Agile SCRUM data engineering team, managing development tasks through Jira and Confluence, participating in sprint planning, code reviews, and cross-team coordination with BI engineers, DevOps, and analytics teams.

Chubb

Feb 2019 - Nov 2021

Big Data Developer

Whitehouse Station, NJ

- Developed large-scale batch data pipelines on Hadoop (HDFS) to process insurance claims, policy transactions, and underwriting data, supporting analytics used by the Risk Modeling and Actuarial Analytics teams.
- Built data ingestion workflows using Apache Sqoop to import structured datasets from Oracle and SQL Server into the Hadoop ecosystem, enabling centralized storage and large-scale analytics for enterprise insurance reporting.
- Implemented transformation logic using HiveQL and Apache Hive, performing aggregations and data cleansing for policy claims and underwriting datasets used by business intelligence and actuarial reporting teams.

- Developed ETL workflows using Apache Pig to process semi-structured claim and audit log data, enabling faster ingestion and transformation for downstream fraud detection and operational analytics.
- Designed and scheduled batch data workflows using Apache Oozie, coordinating multi-step ingestion, transformation, and reporting pipelines while managing job dependencies and failure recovery processes.
- Implemented log ingestion pipelines using Apache Flume, collecting application and transaction logs from internal systems and storing them in HDFS for fraud investigation and audit analysis.
- Developed custom distributed processing logic using Java-based MapReduce jobs, enabling high-volume transformations that required advanced processing beyond standard Hive or Pig workloads.
- Designed Hive table partitioning and bucketing strategies to improve query performance and reduce execution time for large policy and claims datasets accessed by underwriting and analytics teams.
- Performed data cleansing and deduplication workflows using Hive and Unix shell scripting, improving data quality and consistency across datasets used by actuarial and reporting teams.
- Worked closely with the Underwriting Analytics, Risk Management, and Business Intelligence teams to translate reporting requirements into scalable Hadoop-based data processing solutions.
- Tuned cluster performance using YARN resource management and job configuration optimization, improving cluster utilization and supporting multiple concurrent workloads across the data platform.
- Implemented schema evolution and data format optimization using Avro and Parquet, enabling efficient storage and faster query performance for large insurance data tables.
- Designed a long-term data archival strategy within HDFS, aligning with regulatory insurance data retention requirements and ensuring historical claims datasets remained accessible for audit and compliance reviews.
- Documented ETL workflows, Hadoop job dependencies, and operational runbooks using Confluence, improving knowledge sharing and onboarding for new engineers on the data platform team.
- Worked within an Agile SCRUM development team, managing engineering tasks through Jira, supporting production troubleshooting, and performing root cause analysis for failed Hadoop and Hive batch jobs.

CERTIFICATIONS

- Google Cloud Professional Data Engineer
- Snowflake SnowPro Advanced: Architect
- Apache Airflow by Astronomer: DAG Authoring

EDUCATION

Bachelor's of Science